

Atomistic Modeling

What can we do with it?

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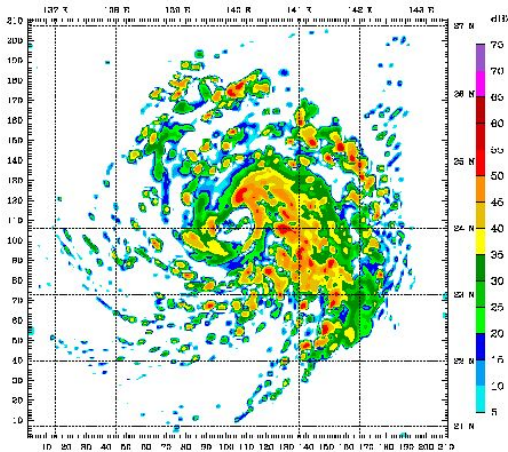
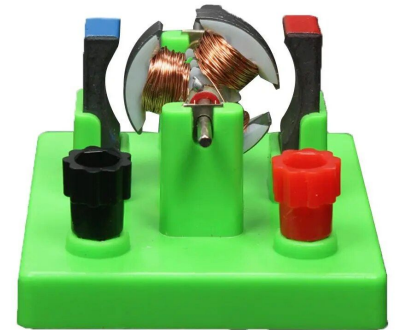
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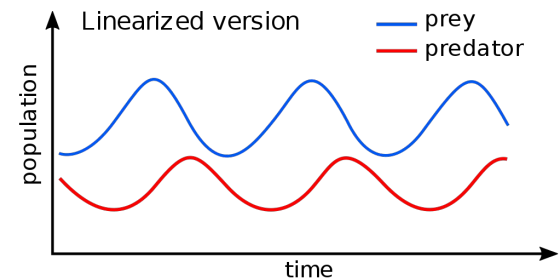
MODELING → Building a **model** of a real system (to study or understand it)

Informative version

“Human” scale
Easier to control
Easier to read



$$\begin{aligned}\frac{dx}{dt} &= \alpha x - \beta xy, \\ \frac{dy}{dt} &= \delta xy - \gamma y,\end{aligned}$$



Models are motivated by the questions you have and the object you are investigating:

- How the atomistic structure of a material look like?
- Is a certain material stable?
- This material is a conductor?
- How this material interact with light?
- Is a molecule polar?
- Where two molecules will bond?
- ...

Models are motivated by the questions you have and the object you are investigating:

$$X_0 \in \mathbb{S} \quad | \quad E(X_0) \leq E(X) \quad \forall X \in \mathbb{S}$$

Structure discover

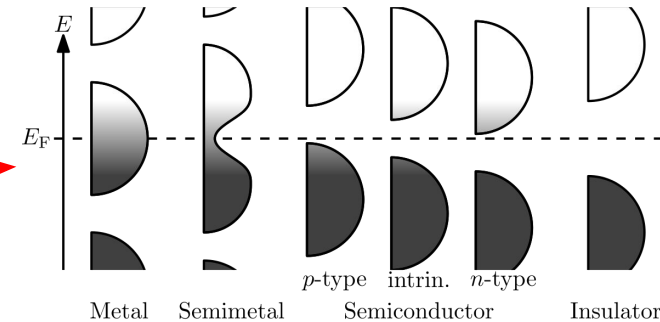
- How the **atomistic structure** of a material look like?
- Is a certain material **stable**?
- This material is a conductor?
- How this material interact with light?
- Is a molecule polar?
- Where two molecules will bond?
- ...

$$\left. \frac{dE}{dX} \right|_{X=X_0} = 0?$$

Some of those questions relate to the potential **energy** of the atomic systems ($E(X_\theta)$) for a certain atomic configuration (X_θ).

Models are motivated by the questions you have and the object you are investigating:

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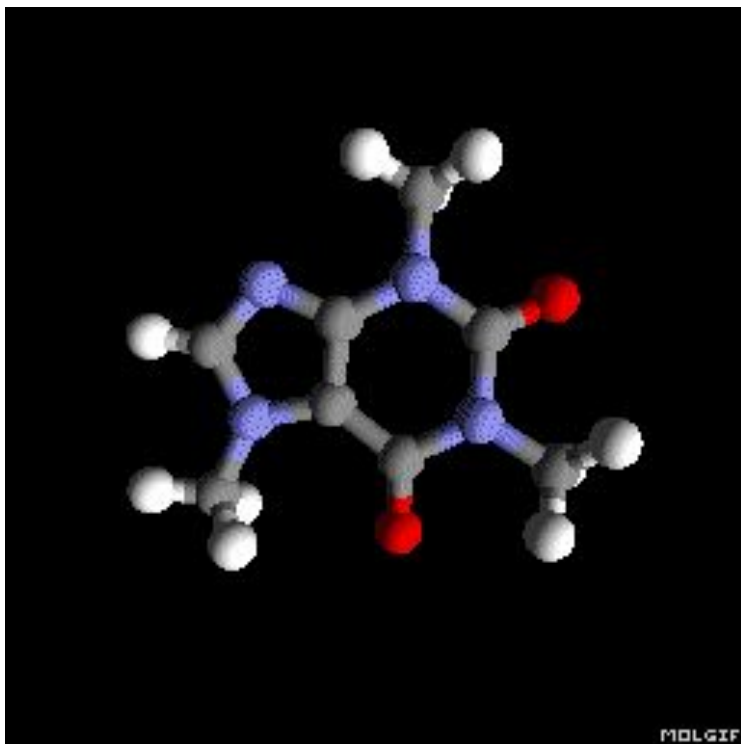


Some of those questions relate to the potential **energy** of the atomic systems ($E(X_\theta)$) for a certain atomic configuration (X_θ).

Others can be answered only by knowing the **electronic** properties of the material, such as the electronic levels of energy (ϵ_i) or the density of states (DOS)

Atomistic modeling: Description of an atomic arrangement in a numerical way and methodologies to compute their energetic and electronic properties.

What is an **atomic configuration**?



We want to inform the computer of the shape and geometric properties of a molecule.

How can we do it?

$$X_0 = \{(S_1, \vec{r}_1), (S_2, \vec{r}_2), \dots, (S_N, \vec{r}_N)\}$$

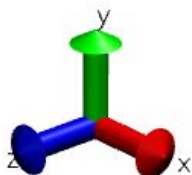
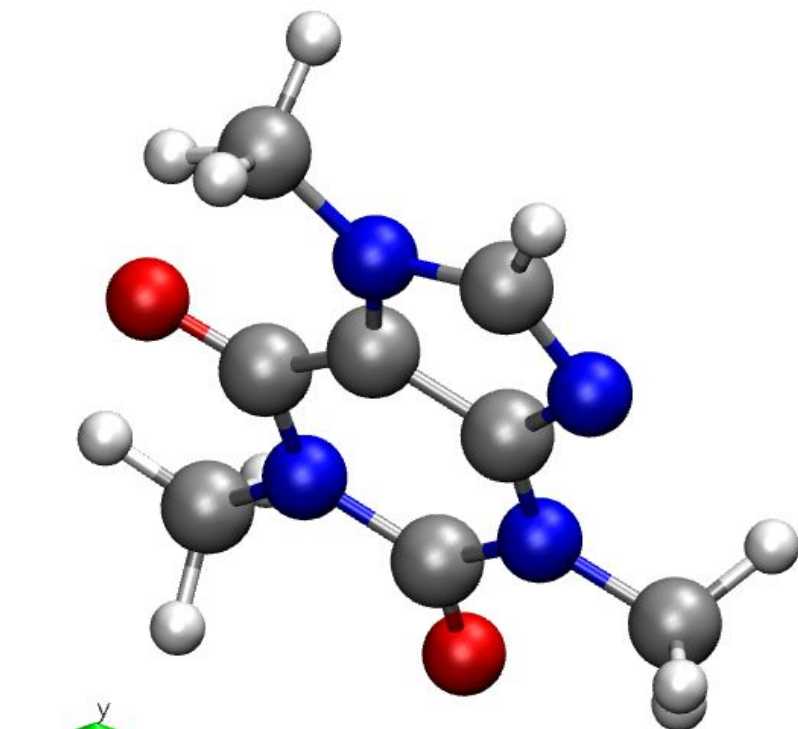


$$\tilde{X}_0 = \left(\begin{bmatrix} S_1 \\ S_2 \\ \dots \\ S_N \end{bmatrix}, \begin{bmatrix} (x_1, y_1, z_1) \\ (x_2, y_2, z_2) \\ \dots \\ (x_N, y_N, z_N) \end{bmatrix} \right)$$

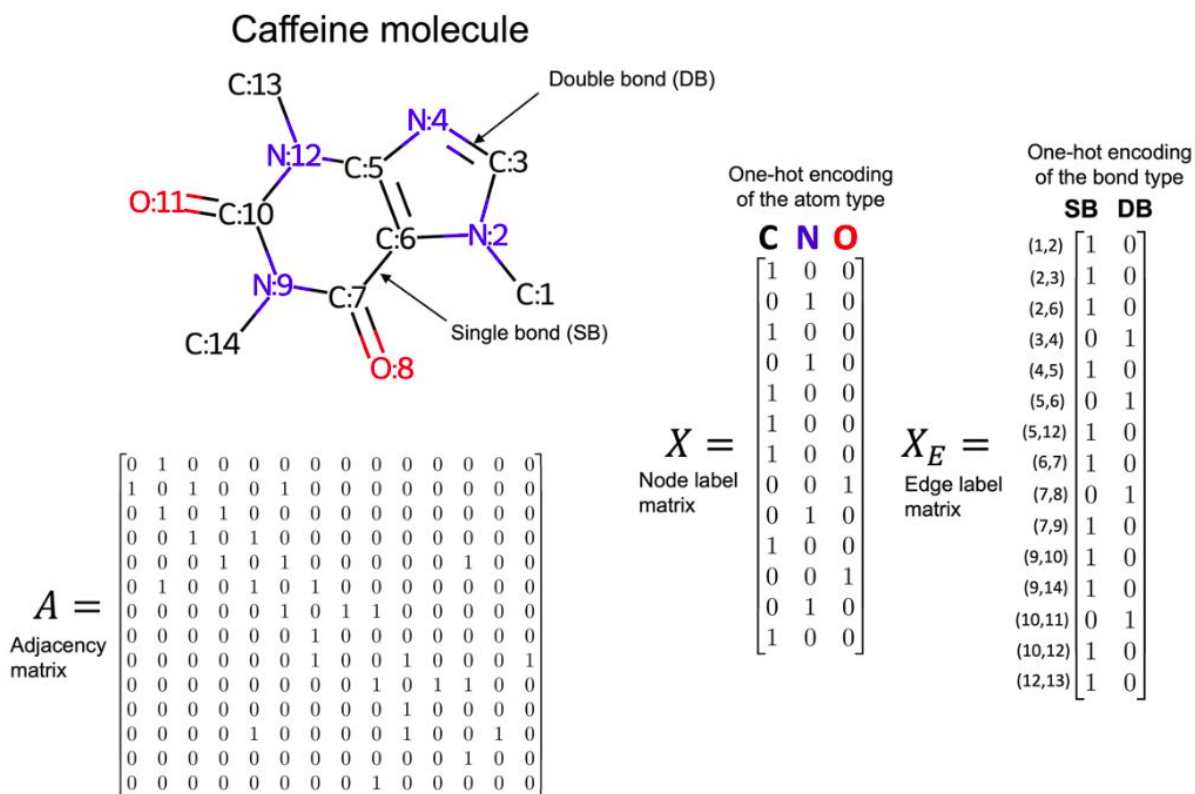


caffeine.xyz

```
24
Caffeine
H      -3.3804130    -1.1272367     0.5733036
N       0.9668296    -1.0737425    -0.8198227
C       0.0567293     0.8527195     0.3923156
N      -1.3751742    -1.0212243    -0.0570552
C      -1.2615018     0.2590713     0.5234135
...
H      -1.2074498     2.7537592     1.7203047
```



Examples of other representations - Graph-based:

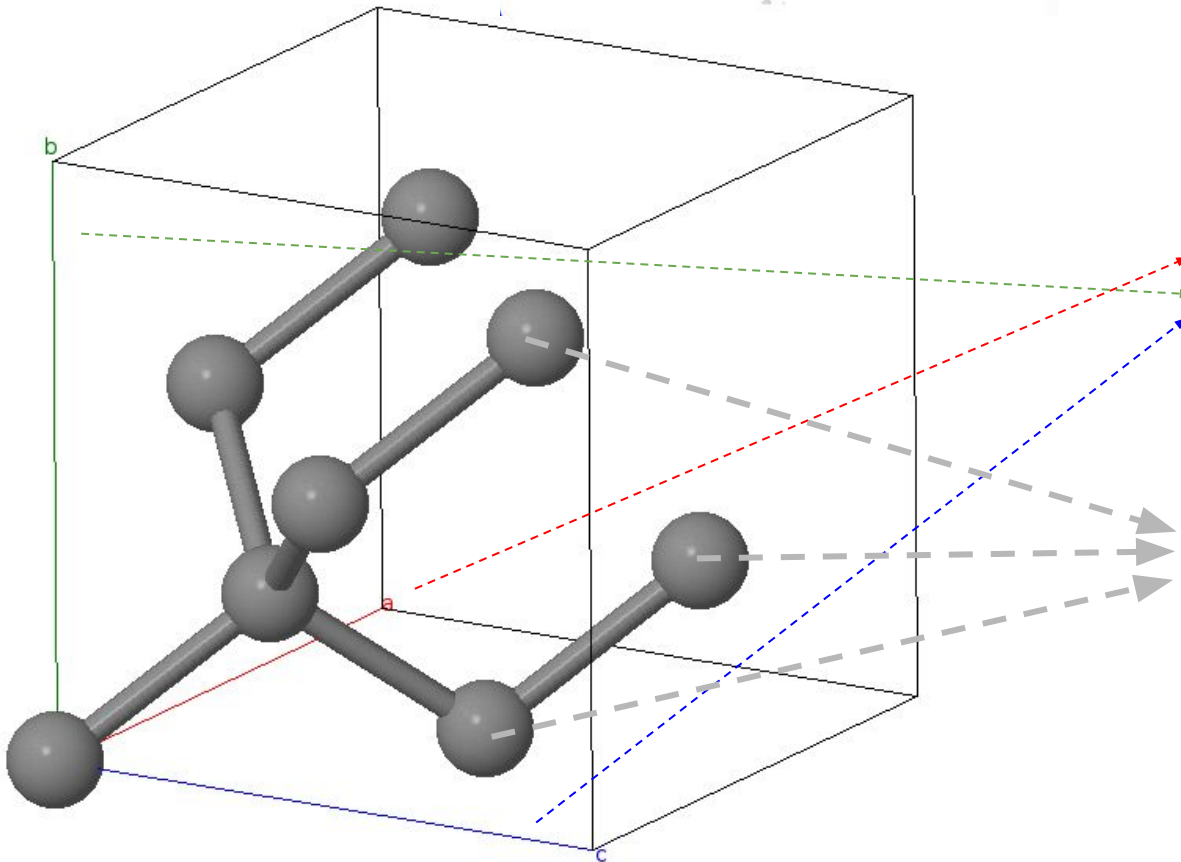


SMILES: CN1C=NC2=C1C(=O)N(C(=O)N2C)C

P 1 [P 1] #1
a=3.460Å
b=3.460Å
c=3.460Å
α=90.000°
β=90.000°
γ=90.000°



With some extra structure, we
can add extra information:

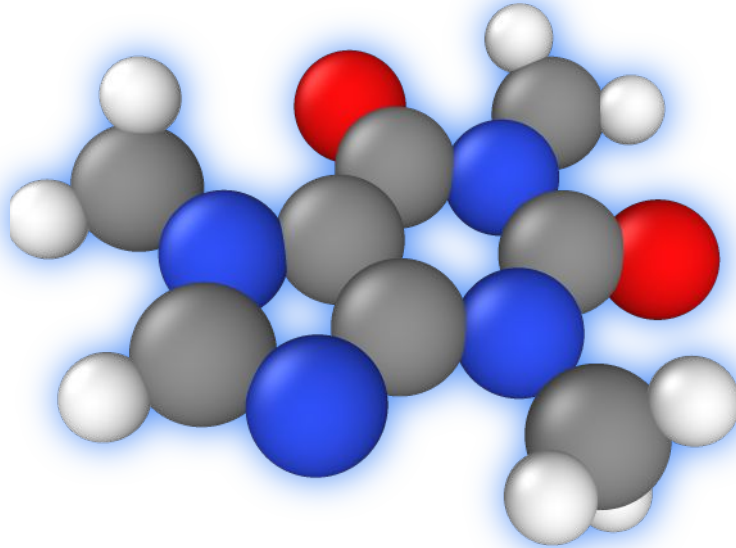


Diamond
1.0
3.46 0.00 0.00
0.00 3.46 0.00
0.00 0.00 3.46
C
8
Direct
0.00 0.00 0.00
0.50 0.50 0.00
0.00 0.50 0.50
0.50 0.00 0.50
0.25 0.25 0.25
0.75 0.75 0.25
0.25 0.75 0.75
0.75 0.25 0.75

POSCAR

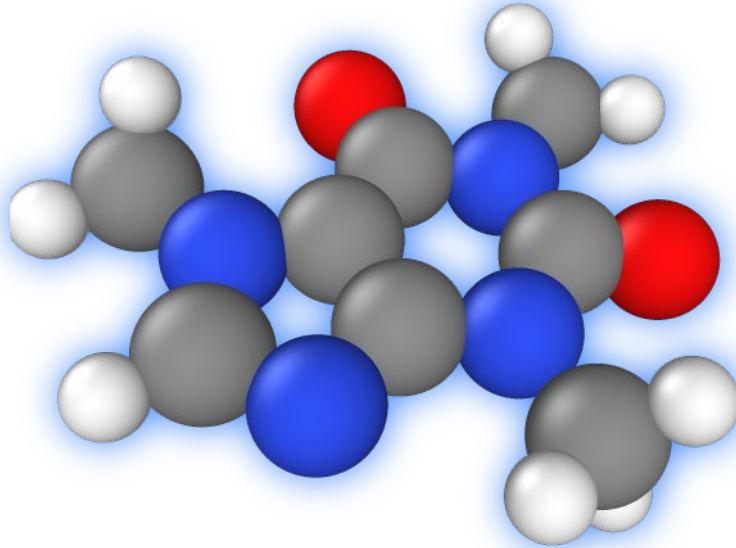
Returning to the problem of $E(X_o)$:

$$X_0 = \{(S_1, \vec{r}_1), (S_2, \vec{r}_2), \dots, (S_N, \vec{r}_N)\}$$



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$$X_0 = \{(S_1, \vec{r}_1), (S_2, \vec{r}_2), \dots, (S_N, \vec{r}_N)\}$$



Classical Potentials:

$$E_{total} = \sum_i U_i \quad \longrightarrow$$

This can always be done, cause the energy is a additive quantity created by the interactions between the atoms. In the other hand, there is no single (or best) way of doing it.

Classical Potentials:

$$E_{total} = \sum_i U_i$$

$$= \Phi_0 + \underbrace{\sum_i \Phi_1(\vec{r}_i)}_{\text{One-body Potential:}} + \underbrace{\sum_{i,j} \Phi_2(\vec{r}_i, \vec{r}_j)}_{\text{Two-body Potential}} + \underbrace{\sum_{i,j,k} \Phi_3(\vec{r}_i, \vec{r}_j, \vec{r}_k) + \dots}_{\text{Three or N-body Potential}}$$

One-body Potential:

Gravitational ($U = \sum m_i \cdot g \cdot h_i$)
 External Electrostatic ($U = \sum q_i V(\mathbf{r}_i)$)

Two-body Potential

Coulomb's Law
 Newtonian gravitation

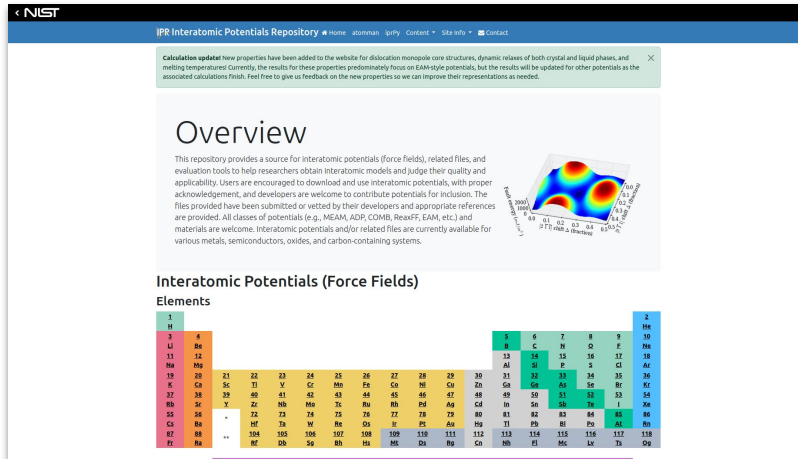
Spring-mass (Hooke)
 Lennard Jones
 Morse

Three or N-body Potential

Tersoff
 TB-SMA
 Universal Potential
 CHARMM
 AMBER

REBO
 AIREBO
 EAM
 MEAN
 ReaxFF

Where to find them?



NIST Interatomic Potentials Repository

Calculation update: New properties have been added to the website for dislocation monolayer core structures, dynamic relaxes of both crystal and liquid phases, and melting temperatures! Currently, the results for these properties predominantly focus on EAM-style potentials, but the results will be updated for other potentials as the associated calculations finish. Feel free to give us feedback on the new properties as we can improve their representations as needed.

Overview

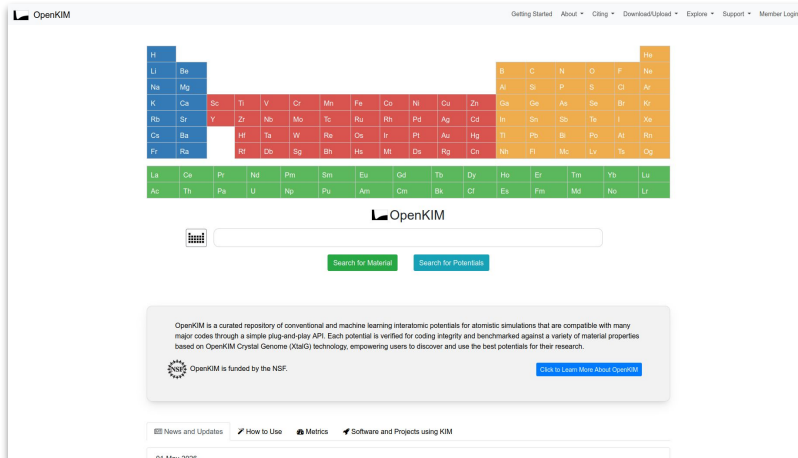
This repository provides a source for interatomic potentials (force fields), related files, and evaluation tools to help researchers obtain interatomic models and judge their quality and applicability. Users are encouraged to download and use interatomic potentials, with proper acknowledgement, and developers are welcome to contribute potentials for inclusion. The files provided have been submitted or vetted by their developers and appropriate references are provided. All classes of potentials (e.g., MEAM, ADP, COMB, ReaxFF, EAM, etc.) and materials are welcome. Interatomic potentials and/or related files are currently available for various metals, semiconductors, oxides, and carbon-containing systems.

Interatomic Potentials (Force Fields) Elements

1	2																	3	4				
H	He																	Li	Be				
3	4																	5	6	7	8	9	10
Li	Be																	B	C	N	O	F	Ne
11	12																	13	14	15	16	17	18
Na	Mg																	Al	Si	P	S	Cl	Ar
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36						
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr						
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54						
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe						
55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72						
Cs	Ba	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu							
87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104						
Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Mn	No	Lr							

IPPR

<https://www.ctcms.nist.gov/potentials/>



OpenKIM

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OpenKIM is a curated repository of conventional and machine learning interatomic potentials for atomic simulations that are compatible with many major codes through a simple plug-and-play API. Each potential is verified for coding integrity and benchmarked against a variety of material properties based on OpenKIM Crystal Genome (XtalG) technology, empowering users to discover and use the best potentials for their research.

OpenKIM is funded by the NSF.

[Click to Learn More About OpenKIM](#)

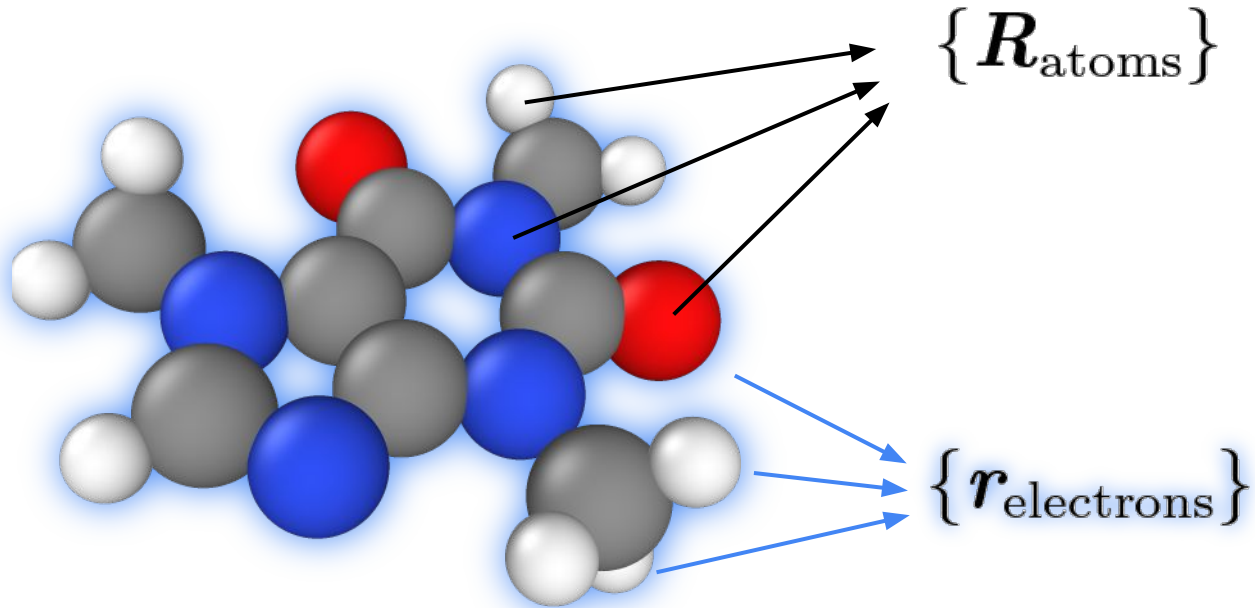
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01 May 2026

OpenKIM

<https://openkim.org/>

Ab initio: The problem is electronic



Given the ondulatory nature of the electron, we will need to address this problem using quantum mechanics. The Hamiltonian (Total Energy) would look like

$$H_{\text{elec.}+\text{atoms}} = \underbrace{-\sum_i \frac{1}{2} \nabla_i^2 - \sum_A \frac{1}{2M_A} \nabla_A^2}_{\text{Kinetic Energy}} - \underbrace{\sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i>j} \frac{1}{r_{ij}} + \sum_{B>A} \frac{Z_A Z_B}{R_{AB}}}_{\text{Potential Energy}}$$

$$H_{\text{elec.}+\text{atoms}} = - \sum_i \frac{1}{2} \nabla_i^2 - \sum_A \frac{1}{2M_A} \nabla_A^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i>j} \frac{1}{r_{ij}} + \sum_{B>A} \frac{Z_A Z_B}{R_{AB}}$$

a proton has ~1840 times the electron mass

Born-Oppenheimer Approximation:

- The use an approximate ansatz for the complete many-body problem that **allows the wave functions of atomic nuclei and electrons to be treated separately.**
- This approximation is motivated by the mass difference between nuclei and electrons.
- The electronic Hamiltonian than can be written as the **kinetic energy of the electrons, the electron interaction with fixed nuclei, and the electron-electron interaction.**

$$\left[- \sum_i^N \frac{1}{2} \nabla_i^2 - \sum_{i,A}^{N, N_{\text{atoms}}} \frac{Z_A}{r_{iA}} + \sum_{i>j}^{N, N} \frac{1}{r_{ij}} \right] \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = E \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$$

$$\left[-\sum_i^N \frac{1}{2} \nabla_i^2 - \sum_{i,A}^{N,N_{\text{atoms}}} \frac{Z_A}{r_{iA}} + \sum_{i>j}^{N,N} \frac{1}{r_{ij}} \right] \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = E \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$$

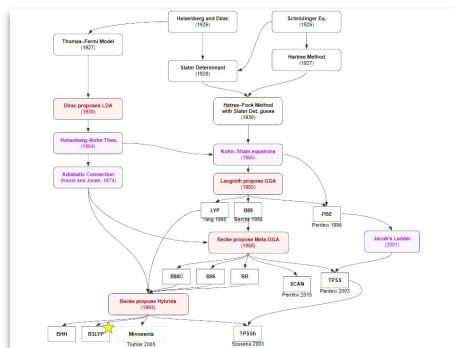
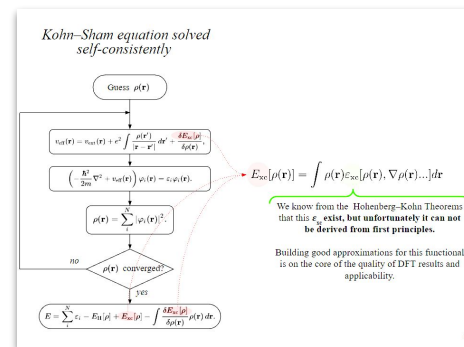
The normalized electronic wavefunction Ψ can be associated to a electronic density $n(\mathbf{r})$:

$$n(\mathbf{r}) = N \int d^3\mathbf{r}' \dots \int d^3\mathbf{r}_N \Psi^*(\mathbf{r}, \dots, \mathbf{r}_N) \Psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$$

This definition **can be inverted for the ground state**, so the density $n_0(\mathbf{r})$ unambiguously defines $\Psi_0[n_0(\mathbf{r})]$, and as consequence defines the ground state energy of the electronic system $E_0[n_0(\mathbf{r})]$.

Hence, **Density Functional Theory!**

Understand the **Kohn–Sham method**, that allow us to numerically solve the DFT and get the electronic properties of atomistic system.

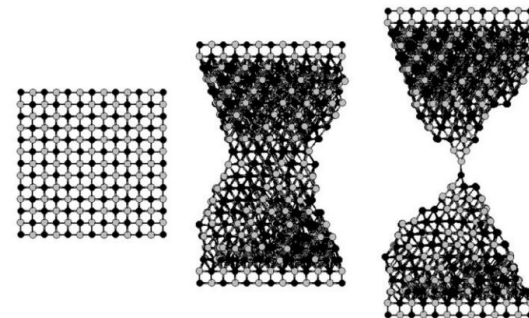
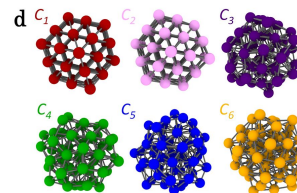
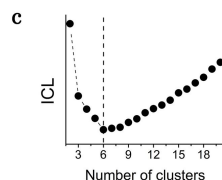
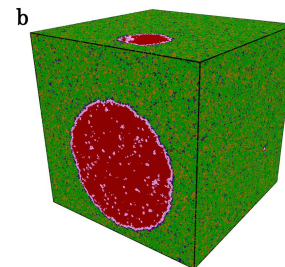
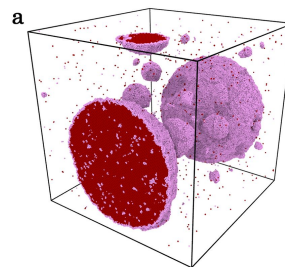
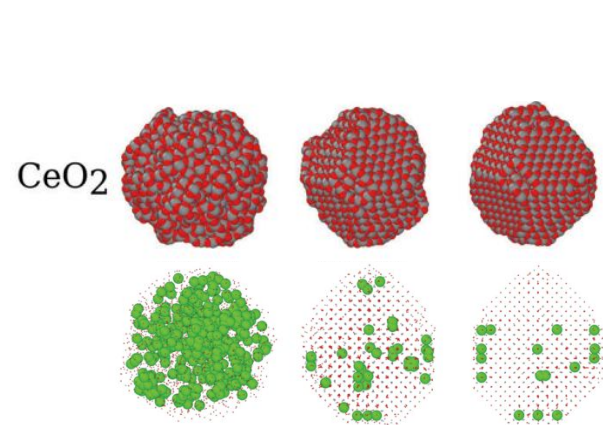
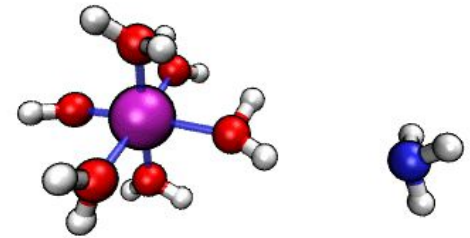


See some of the **Exchange and Correlation Functionals** (for now, “*flavours of DFT*”) available on the literature and their **history**.

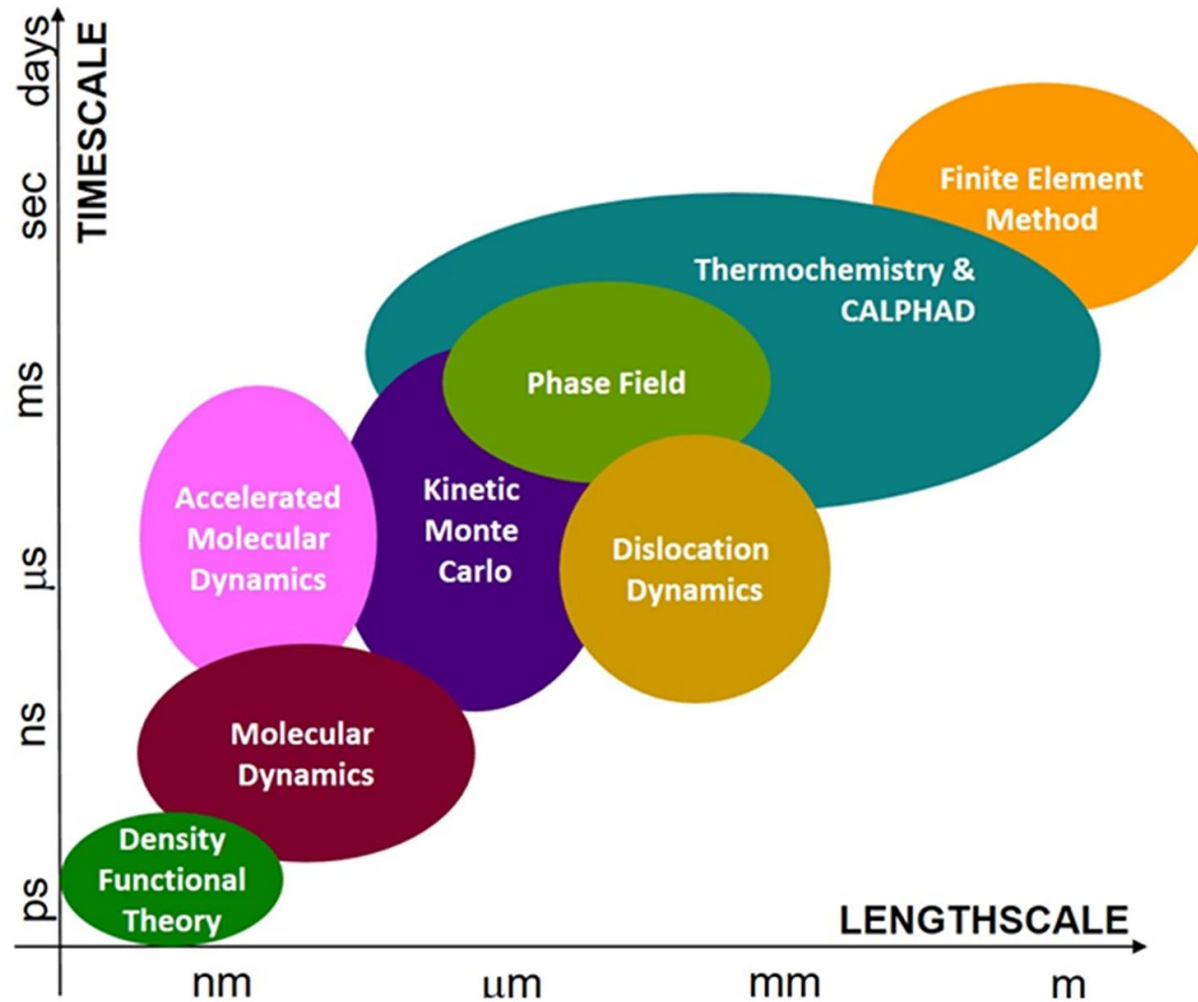
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Molecular Dynamics

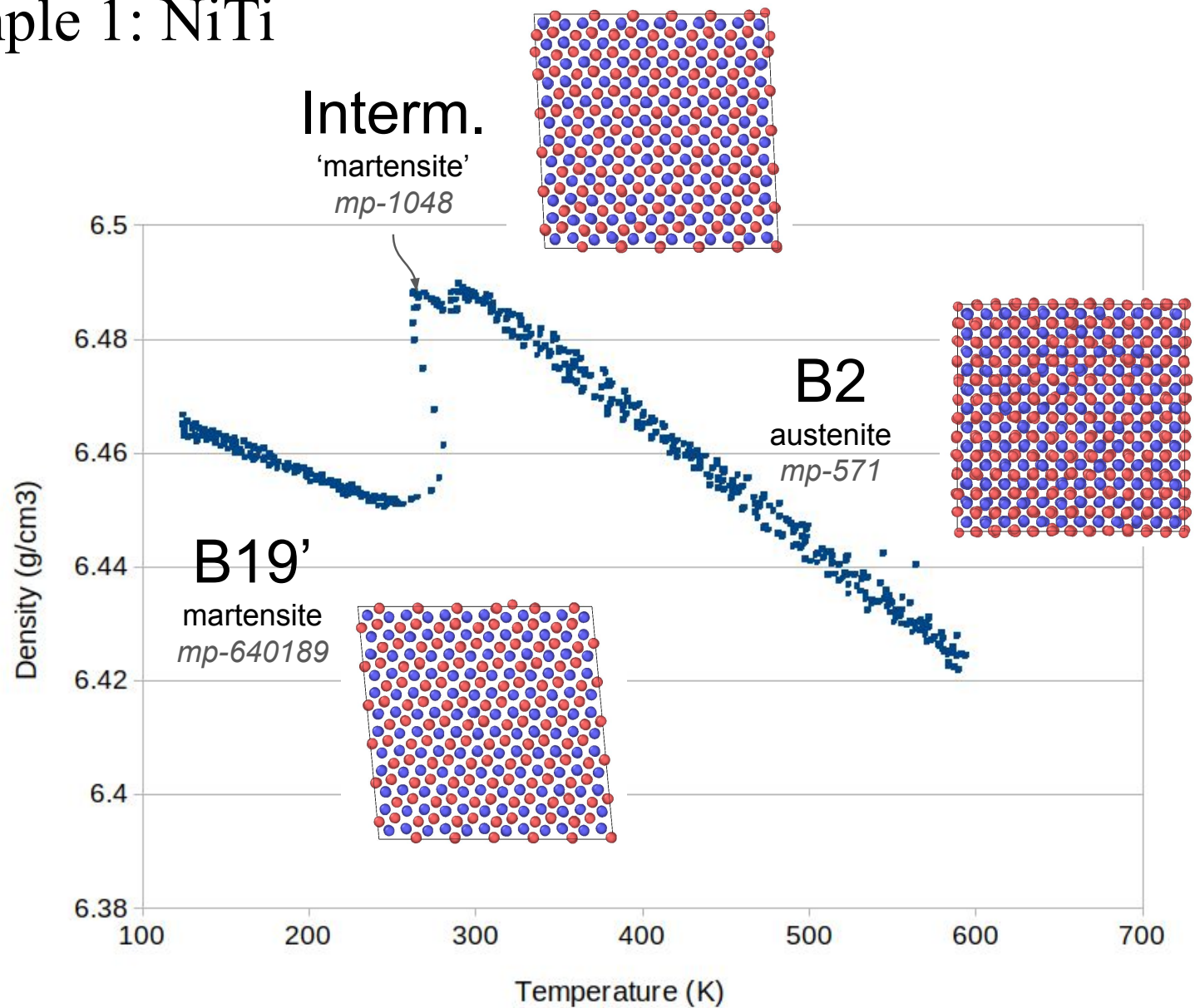
- $\mathbf{F}_i = -\nabla U_i(\mathbf{r}_i) = m_i \mathbf{a}_i = m_i \cdot (d^2 \mathbf{r}_i / dt^2)$
- The forces can be *integrated* via a numerical algorithm (like Velocity Verlet) and we can analyze dynamical phenomena at nanoscale:
 - Chemical reactions
 - Phase transitions
 - Protein folding
 - Battery charging and discharging
 - Gas adsorption
 - Catalysis
 - ...



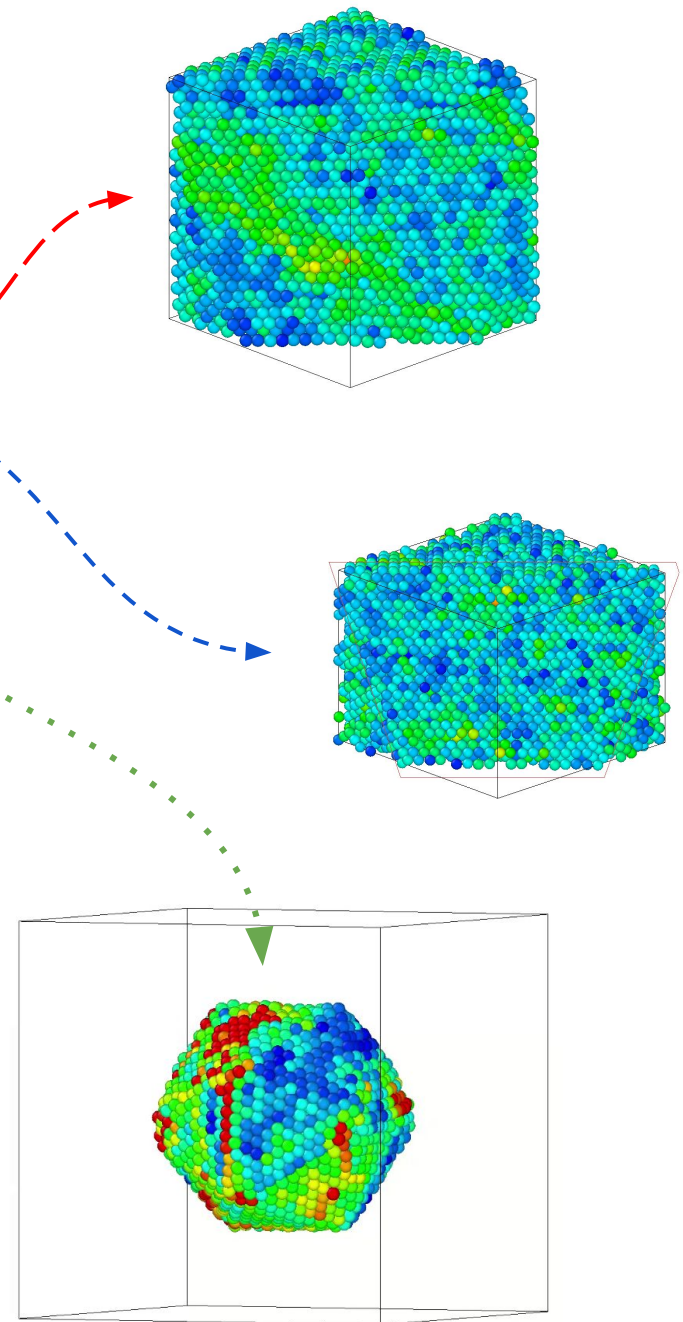
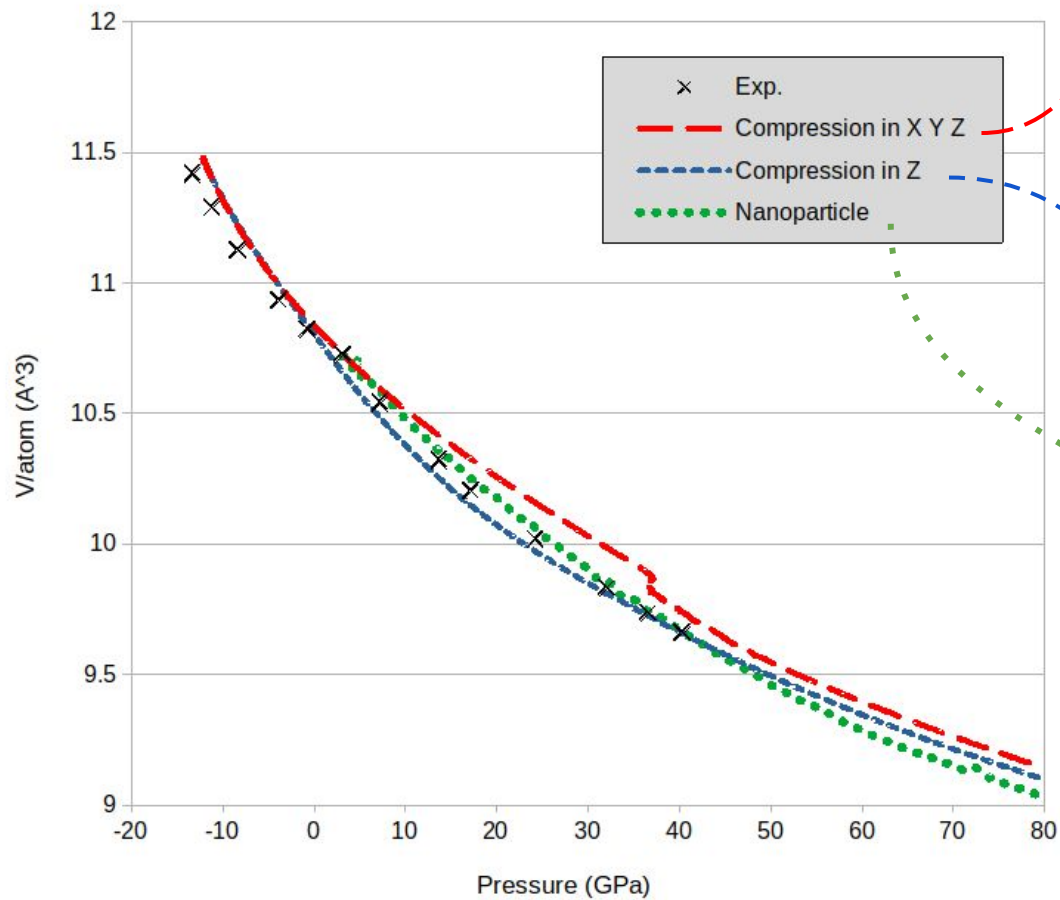
The limits in scale:



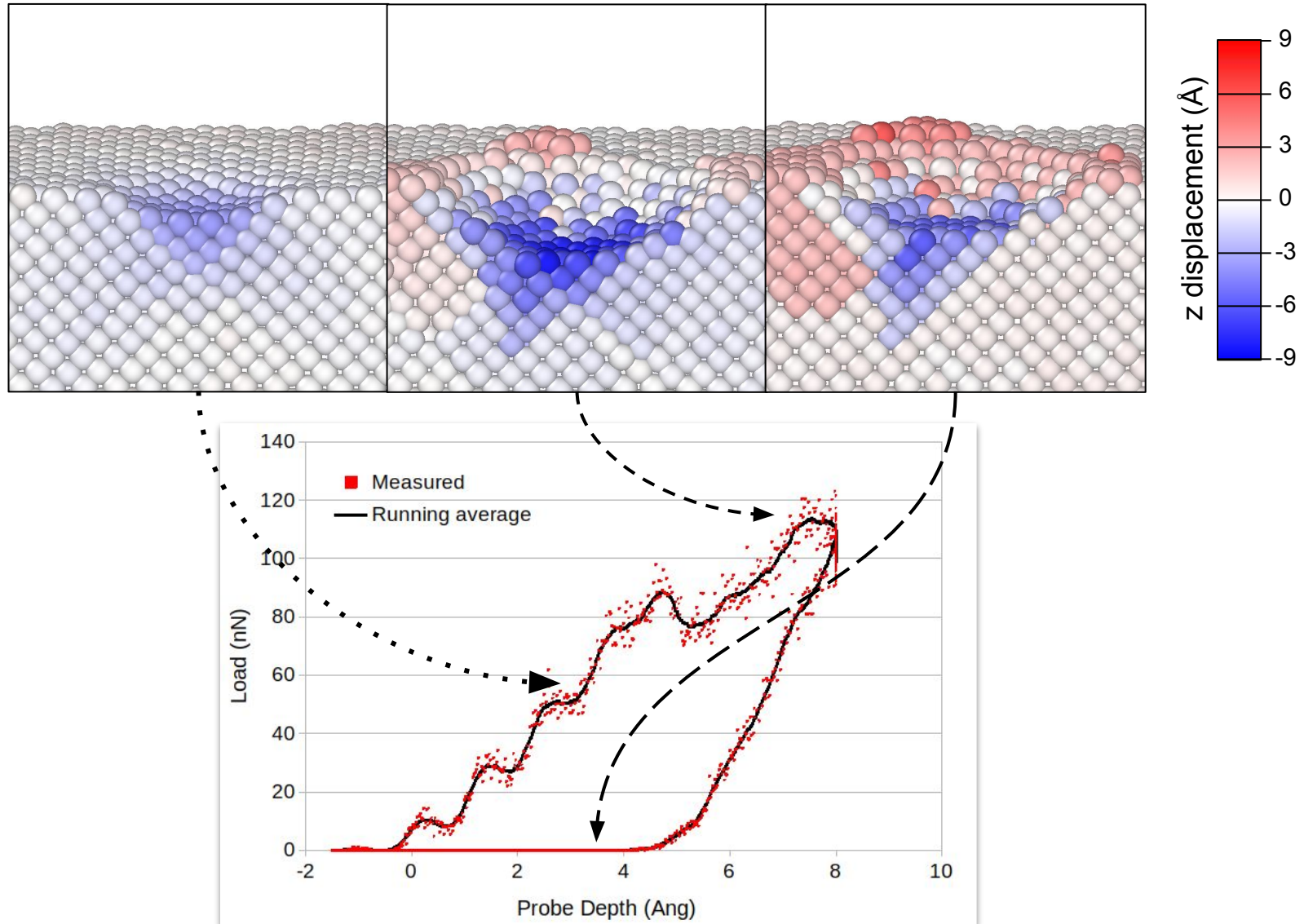
Exemple 1: NiTi



Exemple 2: EoS for CoCrNi



Exemple 2: Nanoindentation of Cu 100 surface



Remainder:

Do you know your system? (Composition, size, structure)

Yes - Create a numerical model (CIF, XYZ, ...)

No - *Structure discover?*

Is electronic structure important?

Yes - You probably will need *ab initio* calculations (DFT, HF, ...)

No - You can do classical potentials, but **ONLY IF** you find a good method

Is temperature or time evolution important?

Yes - You will probably need Molecular Dynamics (i.e., integration of $F=ma$)

No - You can do single point calculations

Thanks for the attention!



Notebook with some MD and DFT examples in Python...

“Here be dragons”

[https://jpalastus.github.io/
joao-paulo.almeida-de-mendonca@grenoble-inp.fr](https://jpalastus.github.io/joao-paulo.almeida-de-mendonca@grenoble-inp.fr)

@alastus

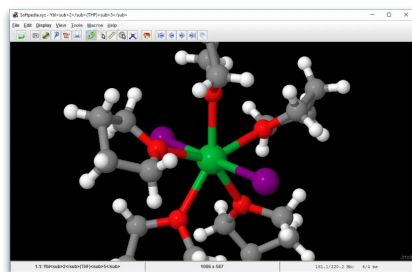
DON'T
FORGET TO
LOVE EACH
OTHER

Supplementary Slides

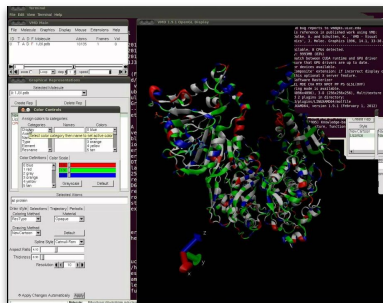
Some homework for the enthusiastic:

Learn to open, visualize and output images from a molecular visualization software or package:

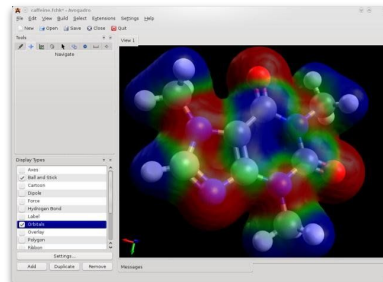
JMol



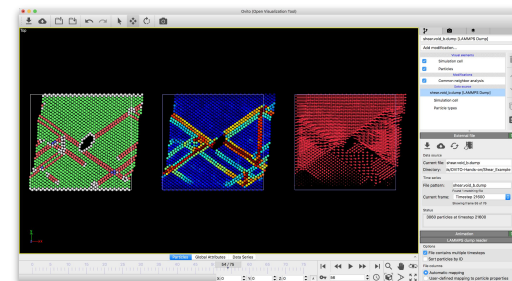
Visual Molecular Dynamics (VMD)



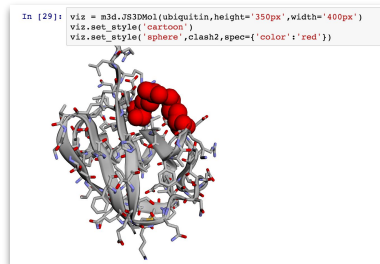
Avogadro



OVITO



```
if pythonist: import py3dmol
```



Or any of the other freeware available for this task!

Try to build some simpler XYZ files by yourself:

